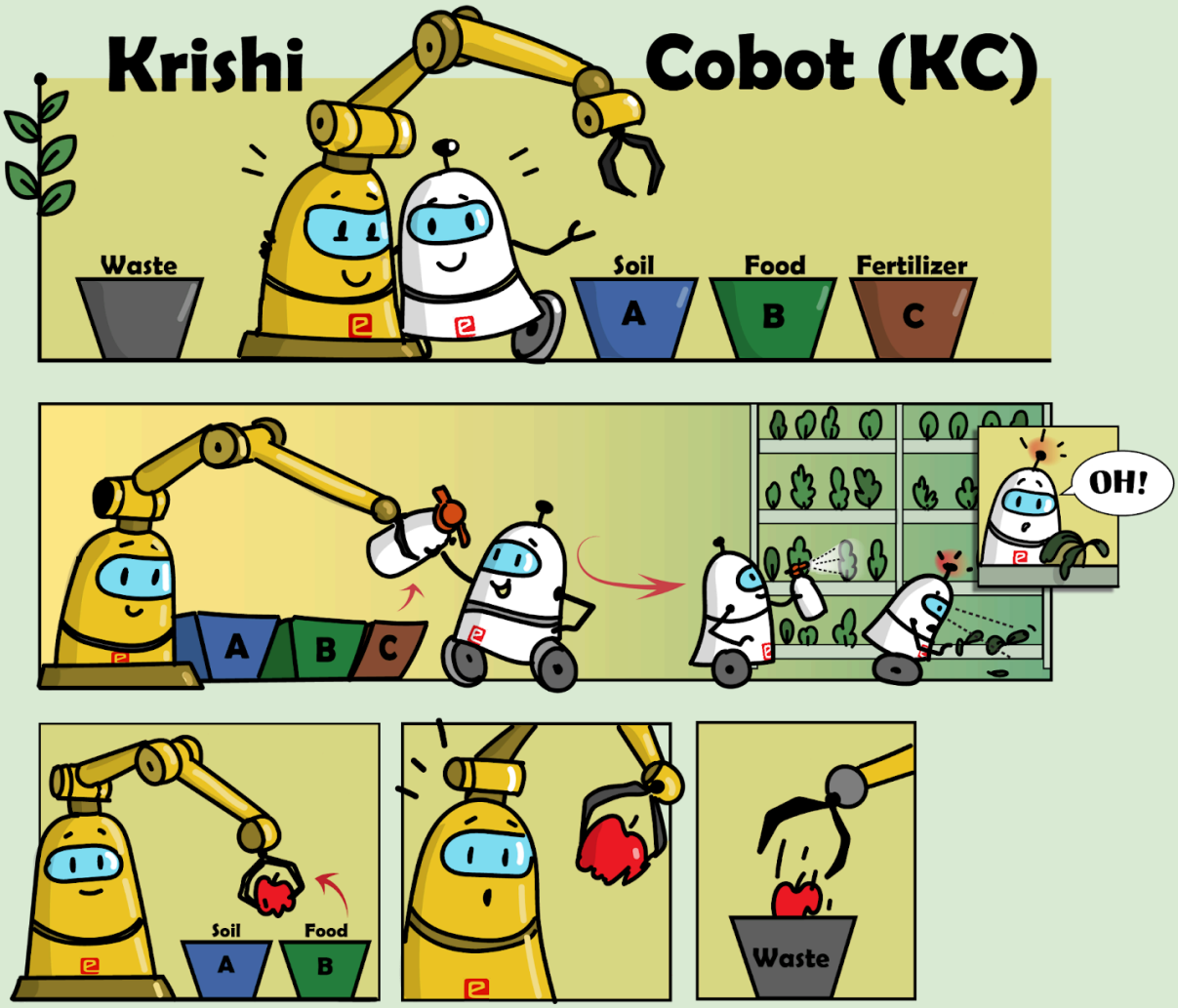
KrishiCobot (KC)

Task 6 Original Configuration



1. Objective:

The goal of this theme is to develop autonomous algorithms for integrated agricultural automation using coordinated mobile robot and robotic arm systems operating in a farmhouse environment.

2. Original Configuration:

a. UR5 Robotic Arm:

i. Manages fertilizer container loading/unloading with the eBot
ii. Inspects fruit trays on conveyor belt, removes bad fruits
iii. Discards bad fruits and empty fertilizer containers in trash bin

b. eBot Mobile Robot:

i. Navigates autonomously between Home Position, Docking Station, and 3 navigation lanes
ii. Receives/delivers fertilizer container from/to UR5
iii. Detects geometric shapes (Triangle/Square/Pentagon) near 8 plant troughs using LiDAR
iv. Publishes plant status with coordinates and plant IDs

3. Scoring Formula :

Total Marks = (600 - T) * 0.05 + (CBI * 1) + (CBD * 5) + (CFI * 1) + (CFL * 5) + (CFU * 5) + (CPD * 5) + (CLN * 2) + (B * 10) - [(CP * 5) + (FDP * 2.5)]

Refer to the rule book to understand the formula